

Rail project controls | technical evidence

Status

DERIVED FROM SIMULATED RECORDS: At 2026-08-03, the GBP 311.6m fictional programme has earned 49.0% of its budgeted scope. CPI 0.906, SPI 0.766, cost variance GBP -15.75m and EAC GBP 343.77m. These are demonstration outputs, not real project performance.

Baseline and ledger design

Baseline assumptions: four sections totaling 68 route-miles, two stations and four signaling installations per section; track GBP1.6m/mile, electrification GBP1.1m/mile, station GBP12m and signal installation GBP2m. Linear time-phased package baselines, 52-week programme, status week30, seed2026. All labour/possession constraints are simplified. Actual physical stations and signal installations are integer-earned; planned fractions can be fractional budget phasing.

Calculation definitions

$PV = \text{sum}(BAC * \text{planned fraction})$; $EV = \text{sum}(BAC * \text{earned physical fraction})$; $AC = \text{sum}(\text{cost})$; $CPI = EV/AC$; $SPI = EV/PV$; $SV = EV - PV$ and $CV = EV - AC$ are currency. $EAC = AC + (BAC - EV)/CPI$, with $AC + \text{unpaid commitments}$ as floor. Remaining commitments cover unspent obligations and are not added to EAC again. No contingency is embedded; risk exposure is reported separately and must not be added mechanically because risks may overlap.

Forecast methods

Baseline completion 2027-01-04; current rate extrapolation UNAVAILABLE: stalled packages; recovery UNAVAILABLE: stalled packages. The recovery assumes 25% more weekly production and 8% premium on remaining forecast cost, giving GBP 357.81m EAC. Latest forecastable package finishes 2027-03-22; stalled packages ['S1-3', 'S1-4', 'S2-3', 'S2-4', 'S3-3', 'S4-3', 'S4-4']. An overall finish is unavailable while any incomplete package has zero recent production. Scenario effectiveness has not been observed. Discrete station/signaling quantities make four-week rates volatile. A zero rate returns no finite finish date.

Limitations

Calendar forecasts extrapolate each package's last four intervals independently; this is not a resource-leveled critical-path schedule, and no statistical confidence interval is claimed. Budget-weighted physical progress is not average miles. SPI becomes uninformative at final completion. ORR four-week periods are not months; moving annual values and missing flags are preserved. Route-km and track-km are distinct; infrastructure breaks preclude naive change claims.

Traceability and sources

OBSERVED CONTEXT: ORR table 3181a supplies 96 railway periods of regional delay rates; table 6320 supplies 79 annual nation records. Context is not used to calibrate simulated project progress. ORR data is licensed OGL v3.0; source Network Rail (infrastructure also TfL and Amey Infrastructure Wales). Source URLs and SHA256

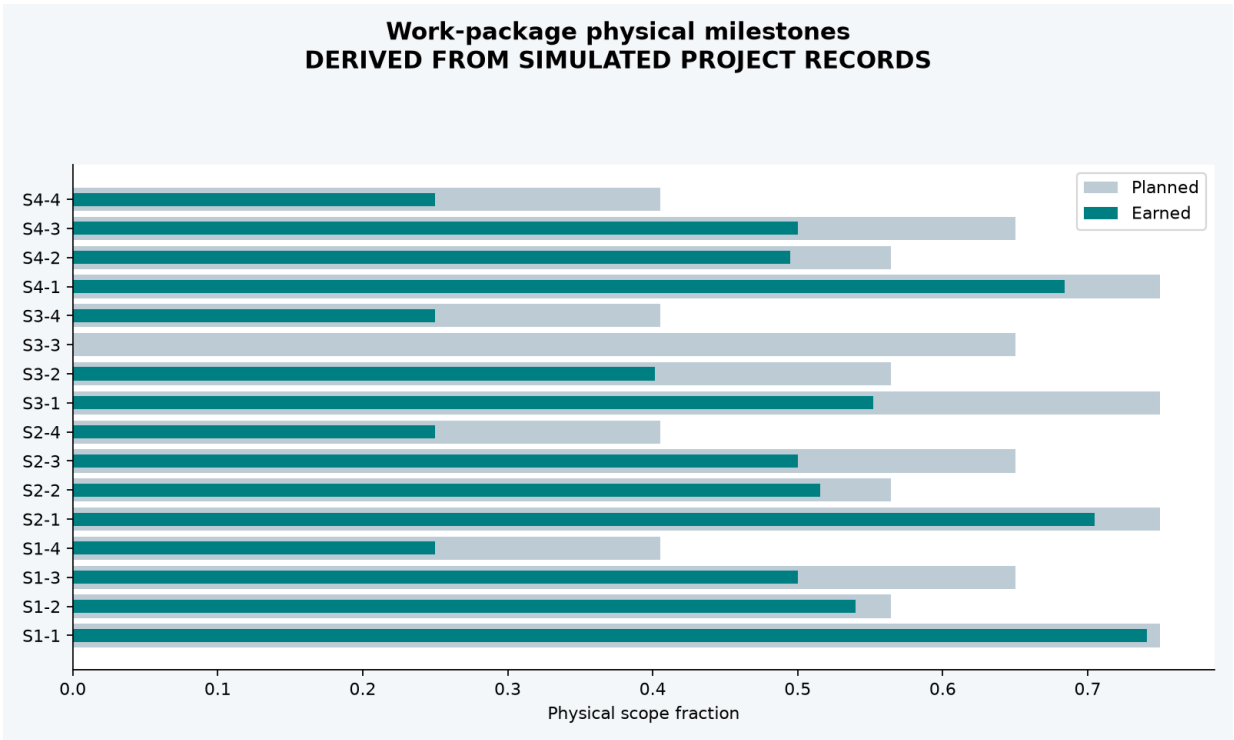
hashes: data/manifest.json.

Source references

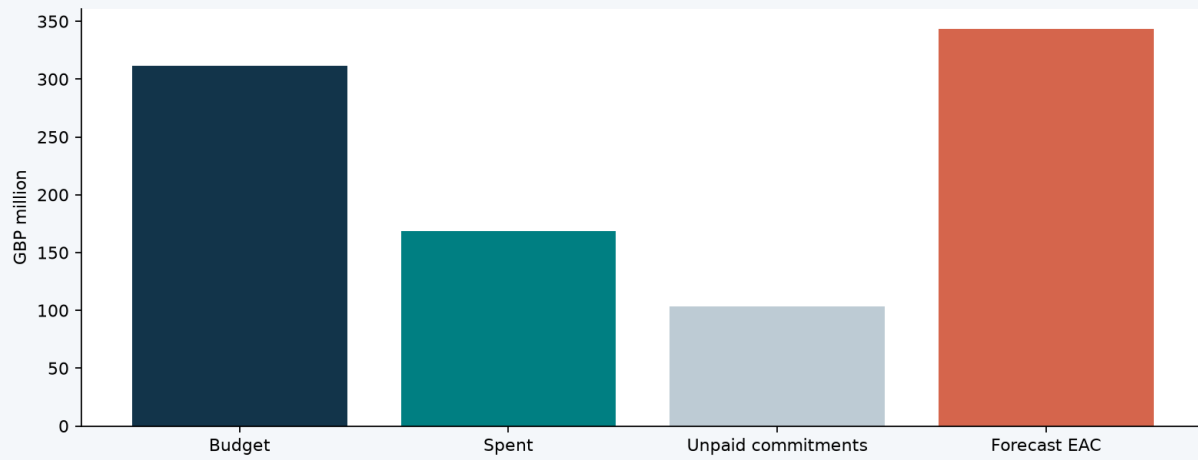
Office of Rail and Road, Table 3181a (accessed 28 September 2026): <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3181a-delay-minutes-per-1-000-train-miles-total-by-network-rail-region-period-1/>

Office of Rail and Road, Table 6320, infrastructure as of March 2025: <https://dataportal.orr.gov.uk/media/1528/table-6320-infrastructure-on-the-mainline.ods>

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Cost outlook; commitments are not added twice DERIVED FROM SIMULATED PROJECT RECORDS



Completion scenario comparison DERIVED FROM SIMULATED PROJECT RECORDS

